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What is Amazon VPC?

Your Virtual Network in the AWS Cloud

Amazon Virtual Private Cloud (VPC) allows you to create a logically isolated virtual network in AWS where you can launch and manage your resources, just like a traditional network in your own data center, but in the cloud.

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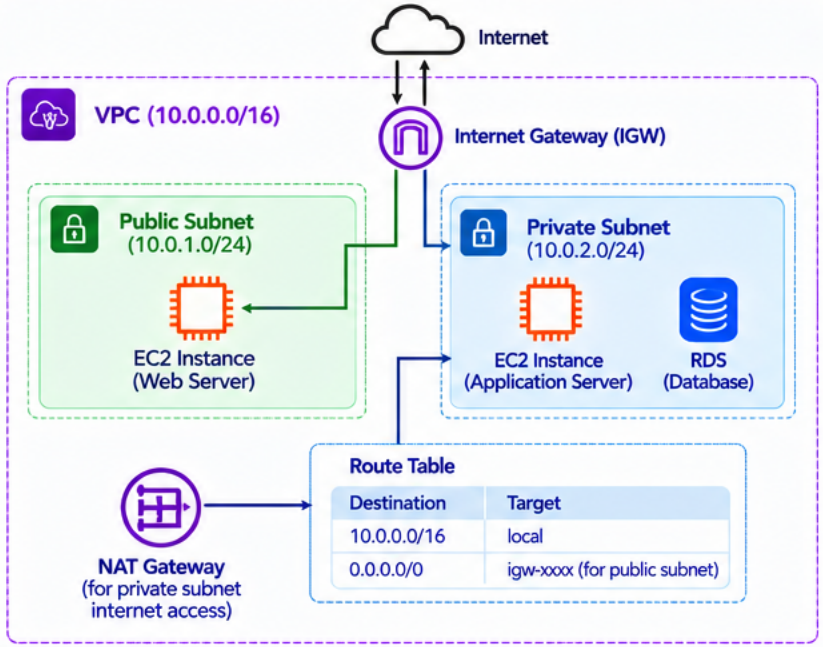
Amazon VPC in Simple Terms

Amazon VPC is a virtual network that gives you full control over your cloud environment. You can define your own IP address range, create subnets, configure route tables, and set up network security, just like you would in a physical data center.

- Isolated and secure network
- You control your IP address range
- Launch AWS resources (e.g., EC2, RDS) inside your VPC
- Connect to the internet or your on-premises network
- Fully customizable networking (environment, routing, security)

Example VPC Architecture



Key Features

Custom IP Range
Choose your own IPv4 and IPv6 CIDR blocks.

Subnets
Divide your VPC into public and private subnets.

Route Tables
Control where network traffic is sent.

Security
Use security groups and network ACLs to control inbound and outbound traffic.

Connectivity
Connect to the internet, on-premises networks (VPN/Direct Connect), or other VPCs.

Scalability
Easily scale your network as your needs grow.

Common Use Cases

- Host web applications
- Run databases in private subnets
- Connect your on-premises data center to AWS
- Create isolated environments for development, testing and production
- Build multi-tier applications
- Meet compliance and security requirements

Key Takeaways

- Amazon VPC gives you a private, isolated network in AWS.
- You have full control over your network configuration.
- Use public subnets for resources that need internet access.
- Use private subnets for resources that should not be directly accessible from the internet.
- Combine with AWS services like IGW, NAT Gateway, VPN and Direct Connect for secure connectivity.

Why It Matters

- Improves security
- Gives you flexibility and control
- Supports hybrid cloud architectures
- Helps you meet compliance requirements
- Builds a strong foundation for scalable cloud applications

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AWS Subnets

Divide. Isolate. Secure. Organize your network.

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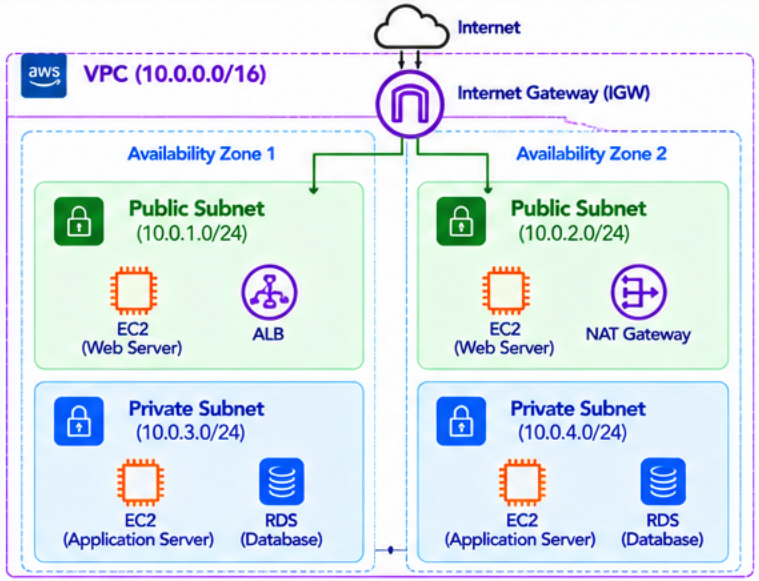
A subnet is a range of IP addresses within your VPC. Subnets allow you to segment your network, place resources in different locations, and control how they can access the internet and other resources.

What is a Subnet?

A subnet (subnetwork) is a smaller range of IP addresses within your VPC. It allows you to divide your VPC into smaller, more manageable networks and control where your AWS resources are placed.

- Part of a VPC (uses the VPC CIDR block)
- Has its own range of IP addresses
- Resides in a specific Availability Zone (AZ)
- Can be public or private
- Helps with security, high availability and better network organization

Example: Subnets in a VPC



Types of Subnets

Public Subnet

- Has a route to the Internet Gateway (IGW)
- Resources can access the internet
- Commonly used for: web servers, load balancers, bastion hosts
- Still uses security groups and network ACLs for control

Private Subnet

- No direct route to the Internet Gateway (IGW)
- Resources cannot be accessed directly from the internet
- Commonly used for: application servers, databases, internal services
- Can access the internet using a NAT Gateway (for outbound traffic)

Subnet CIDR Example

If your VPC has the CIDR block **10.0.0.0/16**, you can create subnets like:

Subnet Name	CIDR Block	Type	Availability Zone
Public Subnet 1	10.0.1.0/24	Public	us-east-1a
Public Subnet 2	10.0.2.0/24	Public	us-east-1b
Private Subnet 1	10.0.3.0/24	Private	us-east-1a
Private Subnet 2	10.0.4.0/24	Private	us-east-1b

Key Points to Remember

- A subnet is created within a VPC.
- Each subnet is in one Availability Zone (AZ).
- Subnets have a range of IP addresses from the VPC CIDR.
- Use public subnets for internet-facing resources.
- Use private subnets for internal resources.
- Route tables control the traffic flow for each subnet.
- Subnets help you build secure, highly available and scalable architectures.

Common Use Cases

- Host web applications (public subnets)
- Run databases (private subnets)
- Application servers (private subnets)
- Bastion hosts / jump servers (public subnets)
- Application Load Balancers (public subnets)
- Isolate environments (dev, test, production)

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AWS Route Tables

Control the flow of traffic in your VPC

A route table contains a set of rules, called routes, that determine where network traffic is directed within your VPC and beyond.

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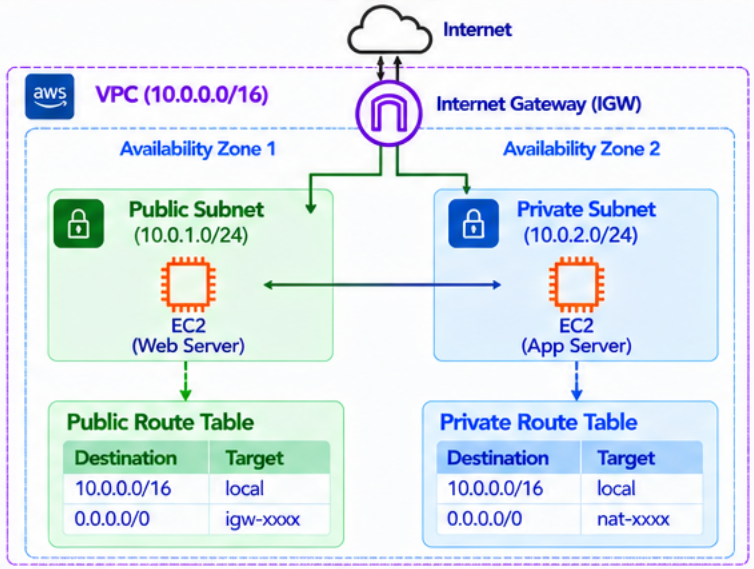
What is a Route Table?

A route table is a set of rules that determine where network traffic is sent for resources in your VPC. Each subnet in a VPC must be associated with a route table, which controls the routing for the subnet.

- Contains a set of routes (rules)
- Determines where network traffic is sent
- Associated with subnets
- Each subnet must be associated with one route table (can be the main route table)
- Helps control access to the internet, on-premises networks and other AWS resources

Route Tables in a VPC (Example Architecture)



Key Components

Routes
Rules that specify the destination (CIDR block) and the target (where to send the traffic).

Route Table
A collection of routes. Each subnet is associated with one route table.

Association
Links a subnet to a route table.

Main Route Table
Automatically created for each VPC. If a subnet is not explicitly associated with a route table, it uses the main route table.

Common Route Table Targets

Internet Gateway (IGW)
For internet access.

NAT Gateway
For outbound internet access from private subnets.

VPC Peering Connection
For traffic to another VPC.

Virtual Private Gateway (VGW)
For on-premises networks (via VPN or Direct Connect).

Transit Gateway
For connecting multiple VPCs and networks.

Example Route Tables

Public Route Table (rtb-xxxx)

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxx

Used by public subnets. Sends internet-bound traffic to the Internet Gateway (IGW).

Private Route Table (rtb-yyyy)

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	nat-xxxx

Used by private subnets. Sends internet-bound traffic to a NAT Gateway.

How Route Tables Work (Traffic Flow Example)

1. An EC2 instance in a **public subnet** sends traffic to the internet (e.g., a website).
2. The instance checks its route table.
3. The destination **0.0.0.0/0** matches a route to the IGW.
4. The traffic is sent to the Internet Gateway and then to the internet.
5. For a private subnet, the route table sends internet-bound traffic to a NAT Gateway instead.

Common Use Cases

- Allow public subnets to access the internet.
- Allow private subnets to access the internet via NAT Gateway.
- Connect your VPC to on-premises networks.
- Connect multiple VPCs using VPC peering or Transit Gateway.
- Control access to specific networks and resources.

Key Takeaway Route tables give you control over where network traffic goes. By associating the right route table with each subnet, you can build secure and flexible network architectures in AWS.

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AWS Internet Gateway (IGW)

Connect your VPC to the Internet

An Internet Gateway (IGW) is a highly available, horizontally scaled AWS service that allows communication between your VPC and the internet.

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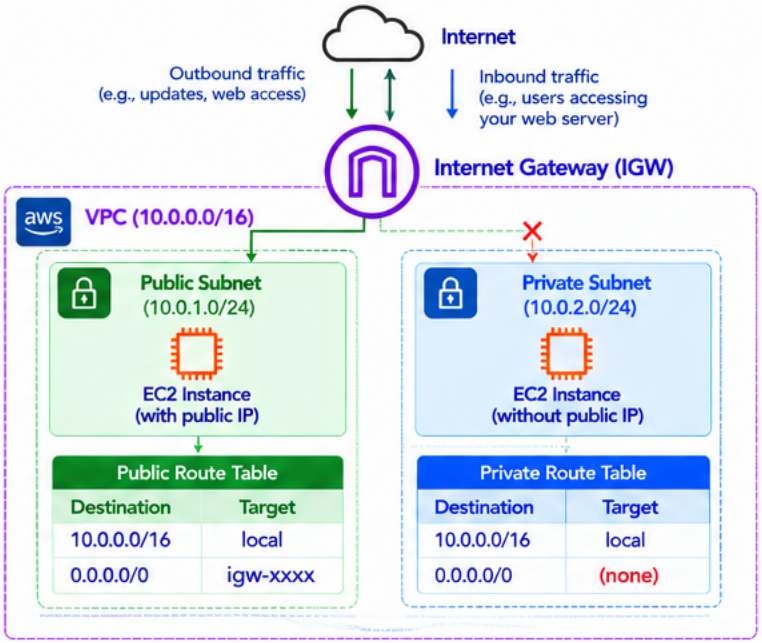
What is an Internet Gateway?

An Internet Gateway (IGW) is a VPC component that enables resources in your VPC to communicate with the internet. It provides a target in your route tables for internet-bound traffic and performs network address translation (NAT) for instances with public IP addresses.

- Horizontally scaled and highly available
- Allows communication between your VPC and the internet
- Provides a target for internet-bound traffic in route tables
- Performs 1:1 NAT for instances with public IP addresses
- You do not manage or scale it (AWS managed)
- One IGW per VPC

How an Internet Gateway Works



Key Features

Highly Available
Horizontally scaled across multiple Availability Zones.

Managed by AWS
You do not create, manage or scale it.

Enables Internet Access
Allows communication between your VPC and the internet.

Performs NAT
Automatically performs 1:1 NAT for instances with public IP addresses.

One per VPC
Each VPC can have only one Internet Gateway.

Common Use Cases

- Host public-facing web applications (e.g., websites, APIs)
- Allow instances to download software updates
- Allow outbound internet access for private resources (via NAT Gateway)
- Enable users on the internet to access resources in your public subnets (e.g., web servers, load balancers)

Steps to Use an Internet Gateway

1. **Create an Internet Gateway**
   aws ec2 create-internet-gateway
2. **Attach it to your VPC**
   aws ec2 attach-internet-gateway –-vpc-id vpc-xxxx
3. **Update your route table**
   Add a route (0.0.0.0/0) with the IGW as the target.
4. **Ensure your resources have public IP addresses**
   Launch instances in public subnets or assign Elastic IPs.

Key Points to Remember

- An Internet Gateway must be attached to a VPC before it can be used.
- Update the route table of public subnets to send internet-bound traffic to the IGW.
- Resources in private subnets cannot access the internet directly (use a NAT Gateway).
- The IGW enables both inbound and outbound internet traffic.
- It is automatically redundant and highly available.

Best Practice

- Use public subnets only for resources that need internet access.
- Keep resources that do not need direct internet access in private subnets.
- Use a NAT Gateway for outbound internet access from private subnets.
- Apply security groups and network ACLs to control traffic.

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AWS NAT Gateway

Secure outbound internet access for your private resources

A NAT Gateway (Network Address Translation Gateway) allows instances in a private subnet to access the internet (for updates, package downloads, etc.) without allowing inbound connections from the internet.

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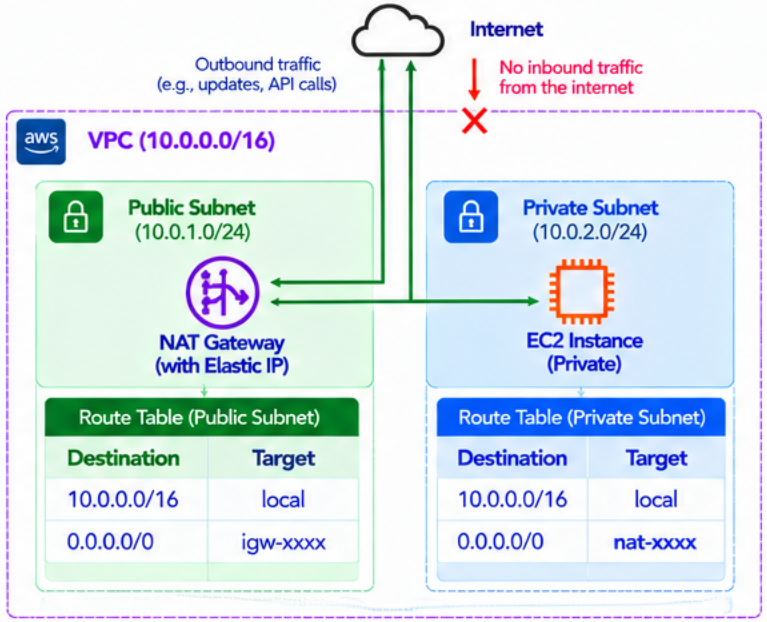
What is a NAT Gateway?

A NAT Gateway is a managed AWS service that enables instances in a private subnet to initiate outbound traffic to the internet while preventing inbound traffic from the internet.

- Provides outbound internet access for private subnets
- Blocks inbound connections from the internet
- Managed by AWS (no servers to manage)
- Highly available within an Availability Zone
- Uses Elastic IP address (public IP)
- Automatically scales
- Ideal for private instances that need to download updates, access APIs, etc.

How a NAT Gateway Works (Architecture)



Key Components

NAT Gateway
A managed service that performs network address translation (NAT) for instances in private subnets.

Elastic IP
A public IP address that is associated with the NAT Gateway.

Public Subnet
The subnet where the NAT Gateway resides. Must have a route to the Internet Gateway (IGW).

Private Subnet
The subnet with instances that need outbound internet access.

Route Tables
Routes in private subnets point 0.0.0.0/0 to the NAT Gateway.

Use Cases

- Download software updates (e.g., yum, apt)
- Access public AWS services (e.g., S3, DynamoDB APIs)
- Pull container images (e.g., from Docker Hub, ECR)
- Call external APIs
- Allow private databases or application servers to access the internet securely

Route Table Examples

Public Subnet Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxx

Private Subnet Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	nat-xxxx

Key Takeaways

- NAT Gateway allows outbound internet access only.
- It does not allow inbound traffic from the internet.
- Deploy it in a public subnet.
- Update private subnet route tables to point to the NAT Gateway.
- It is highly available and automatically scales.
- Use one NAT Gateway per Availability Zone for high availability.
- Consider costs, as you are charged per hour and per GB of data.

Best Practices

- Use one NAT Gateway per Availability Zone.
- Place it in a public subnet.
- Use private subnets for resources that should not be publicly accessible.
- Monitor usage with CloudWatch.
- Review costs and consider alternatives (e.g., VPC Endpoints for AWS services where possible).

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AWS Security Groups

Act as a virtual firewall for your AWS resources

A Security Group is a stateful virtual firewall that controls inbound and outbound traffic to your AWS resources, such as EC2 instances, RDS databases, and load balancers.

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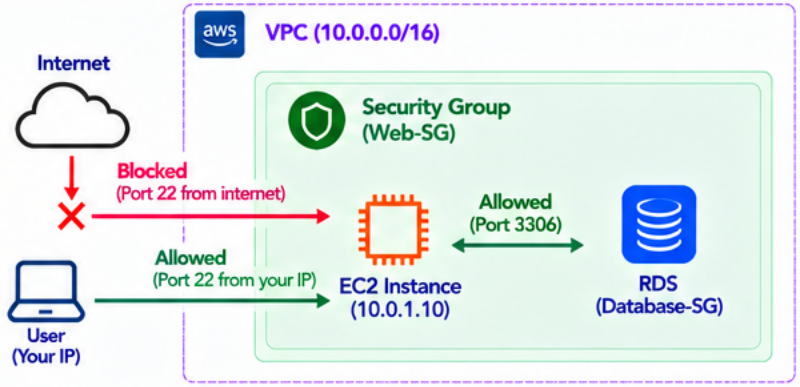
What is a Security Group?

A Security Group (SG) is a stateful virtual firewall that controls inbound and outbound traffic to AWS resources. You define rules that allow or deny traffic based on protocol, port number and source or destination IP addresses (or other security groups).

- Stateful (return traffic is automatically allowed)
- Acts at the instance/resource level
- Supports allow rules only (no explicit deny)
- Applied to resources in a VPC
- You can have multiple security groups per resource
- Uses IP addresses, CIDR ranges or other security groups as sources/destinations

How Security Groups Work (Example)



Inbound Rules (Web-SG)

Type	Protocol	Port	Source
SSH	TCP	22	203.0.113.10/32 (Your IP)
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

Outbound Rules (Web-SG)

Type	Protocol	Port	Destination
All Traffic	All	All	0.0.0.0/0 (Default)

Key Features

- **Stateful**
  If you allow inbound traffic, the response traffic is automatically allowed.
- **Allow Rules Only**
  You specify what to allow. There is no explicit deny rule (everything else is denied by default).
- **Resource Level**
  Applied to ENIs (e.g., EC2, RDS, ALB, etc.).
- **Multiple SGs**
  A resource can be associated with multiple security groups.
- **Works within a VPC**
  Security groups are specific to a VPC and cannot be shared across VPCs.

Common Use Cases

- Allow SSH access to EC2 from your IP only
- Allow HTTP/HTTPS to a web server from the internet
- Allow application servers to connect to a database on port 3306
- Control traffic to an Application Load Balancer
- Restrict access between tiers (e.g., web, app, database)

Security Group Rules Explained

Direction	What it controls	Common examples
Inbound	Traffic coming to the resource (instance, database, etc.)	• Allow HTTP (port 80) from 0.0.0.0/0 • Allow SSH (port 22) from your IP • Allow MySQL (port 3306) from app server SG
Outbound	Traffic going out from the resource	• Allow all traffic (0.0.0.0/0) - default • Allow HTTPS (port 443) to the internet • Allow access to another security group

Key Takeaways

- Security groups act as a firewall for your AWS resources.
- They are stateful and allow return traffic automatically.
- You define allow rules for inbound and outbound traffic.
- You can use IP addresses, CIDR ranges or other security groups as sources/destinations.
- A resource can have multiple security groups.
- Follow the principle of least privilege (allow only what you need).

Best Practices

- Allow only necessary ports and IP ranges.
- Use specific IP addresses instead of 0.0.0.0/0 where possible.
- Use security groups (not IPs) to control traffic between AWS resources.
- Keep rules organized with clear descriptions.
- Review your security group rules regularly.
- Combine with NACLs for an additional layer of security (at the subnet level).

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AWS Network ACLs

Subnet-level firewall for your VPC

A Network ACL (NACL) is a stateless firewall that controls inbound and outbound traffic at the subnet level in your VPC.

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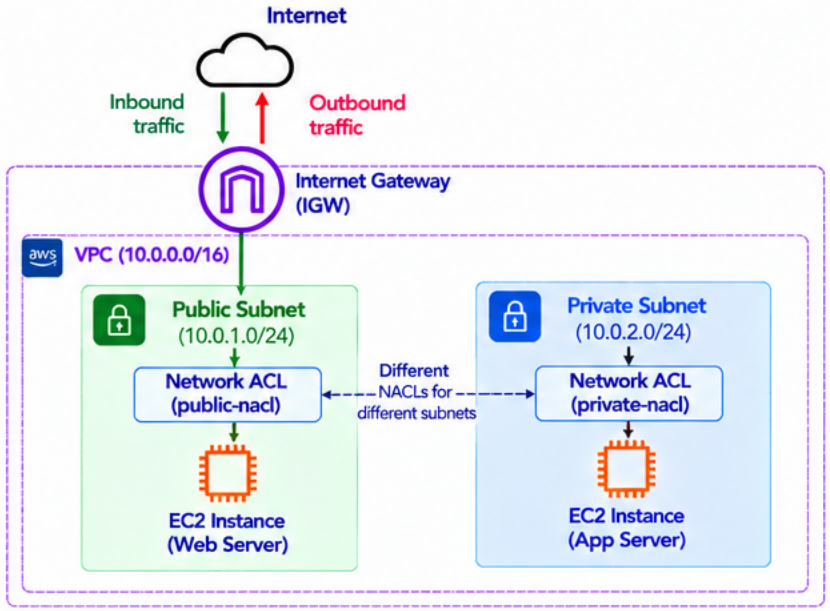
What is a Network ACL?

A Network ACL (NACL) is a stateless firewall that controls inbound and outbound traffic at the subnet level. It allows or deny traffic based on rules that you define, using protocol, port number and source or destination IP addresses.

- Operates at the subnet level
- Stateless (you must allow both inbound and outbound traffic)
- Supports allow and deny rules
- Evaluates rules in number order (from lowest to highest)
- Each subnet is associated with one NACL (can be default or custom)
- Helps provide an additional layer of security in your VPC

How Network ACLs Work (Example)



Key Features

Stateless
You must explicitly allow both inbound and outbound traffic.

Subnet Level
Each subnet is associated with one NACL (default or custom).

Rule Evaluation Order
Rules are evaluated in number order (from lowest to highest).

Allow and Deny Rules
You can explicitly allow or deny traffic.

Default NACL
Every VPC has a default NACL that allows all inbound and outbound traffic.

Common Use Cases

- Add an extra layer of security to subnets
- Block specific IP ranges or ports
- Control traffic between public and private subnets
- Meet compliance and security requirements
- Restrict access to sensitive resources

Example NACL Rules (Public Subnet)

Inbound Rules

Rule #	Type	Protocol	Port Range	Source	Action
100	SSH	TCP	22	0.0.0.0/0	ALLOW
110	HTTP	TCP	80	0.0.0.0/0	ALLOW
120	HTTPS	TCP	443	0.0.0.0/0	ALLOW
200	All	All	All	0.0.0.0/0	DENY

Outbound Rules

Rule #	Type	Protocol	Port Range	Destination	Action
100	All	All	All	0.0.0.0/0	ALLOW
200	All	All	All	0.0.0.0/0	DENY

Note: You must add an outbound rule to allow the return traffic. NACLs are stateless.

NACL vs Security Groups

Feature	Network ACL (NACL)	Security Group (SG)
Level	Subnet	Instance (ENI)
State	Stateless	Stateful
Default	Allows all traffic	Denies all traffic
Rules	Allow and deny	Allow only
Rule order	Number order	No specific order
Scope	One per subnet	Multiple per instance
Use case	Subnet-level control and compliance	Instance-level control

Key Takeaways

- NACLs control traffic at the subnet level.
- They are stateless, so allow both inbound and outbound traffic.
- Use custom NACLs for tighter control.
- Combine NACLs with Security Groups for defense in depth.

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AWS Networking Architecture

A complete and practical VPC design

Connect, secure and scale your applications on AWS

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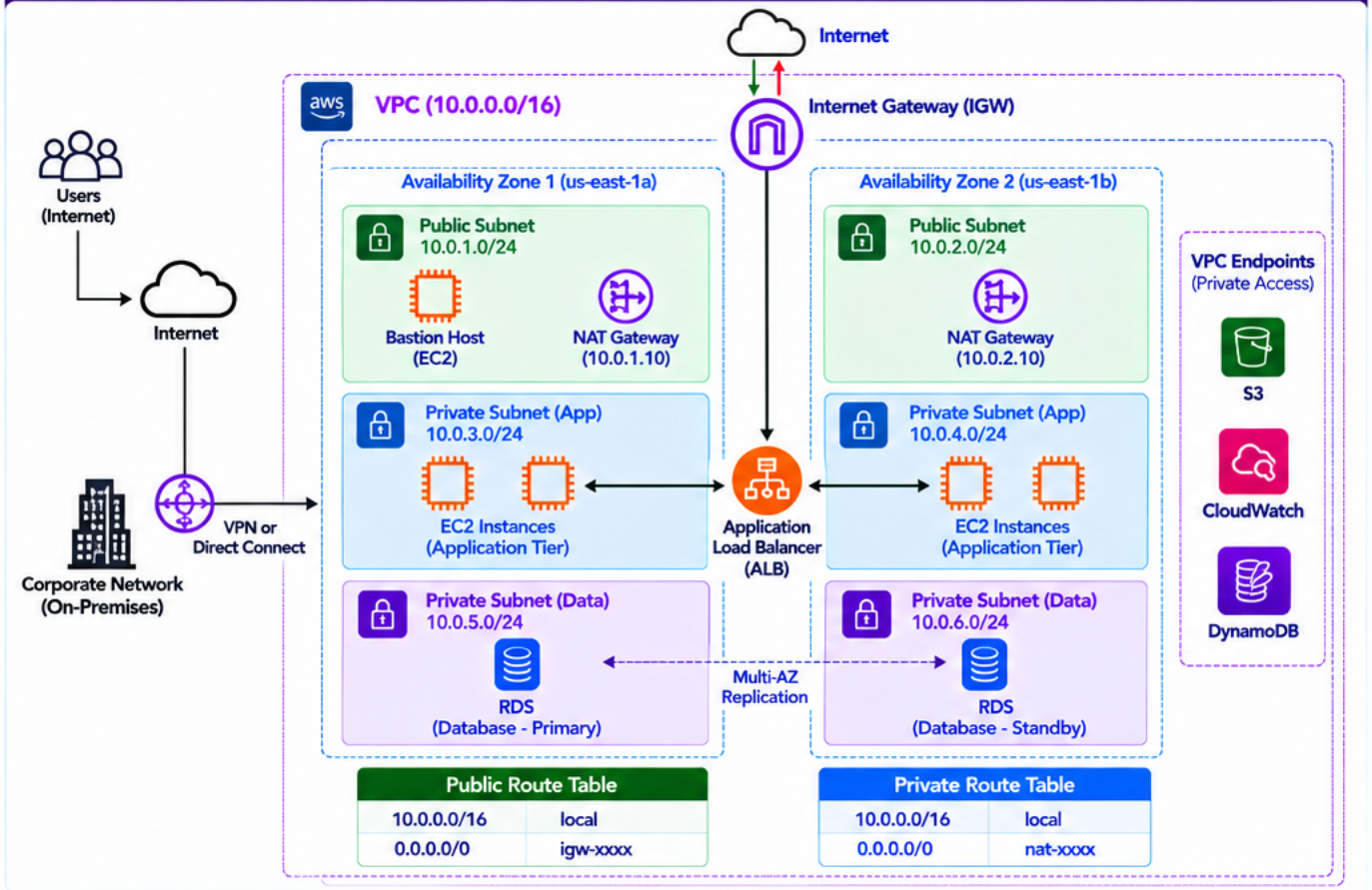
AWS Networking Architecture (Multi-AZ VPC)



Public Route Table	
10.0.0.0/16	local
0.0.0.0/0	igw-xxxx

Private Route Table	
10.0.0.0/16	local
0.0.0.0/0	nat-xxxx

Key Components

VPC
Your isolated virtual network in AWS.

Subnets
Divide your VPC into public and private subnets across multiple Availability Zones.

Internet Gateway (IGW)
Allows communication between your VPC and the internet.

NAT Gateway
Provides outbound internet access for instances in private subnets.

Route Tables
Control where network traffic is sent.

Security Groups
Stateful firewall that controls traffic at the instance level.

Network ACLs
Stateless firewall that controls traffic at the subnet level.

Load Balancer (ALB)
Distributes incoming traffic across multiple instances.

VPC Endpoints
Private access to AWS services without using the internet.

VPN / Direct Connect
Securely connect your on-premises network to your VPC.

Benefits of This Architecture

- High availability across multiple Availability Zones
- Secure and isolated network environment
- Controlled internet access (public and private subnets)
- Scalable and resilient application architecture
- Private access to AWS services

Key Takeaways

- Use public subnets for resources that need internet access (e.g., ALB, bastion, NAT).
- Use private subnets for application and database resources.
- Control traffic with route tables, security groups and network ACLs.
- Use VPC endpoints to access AWS services privately.
- Design across multiple Availability Zones for high availability.

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AWS VPC Peering

Securely connect your VPCs

VPC Peering allows you to connect two VPCs privately over the AWS network, enabling them to communicate as if they are in the same network.

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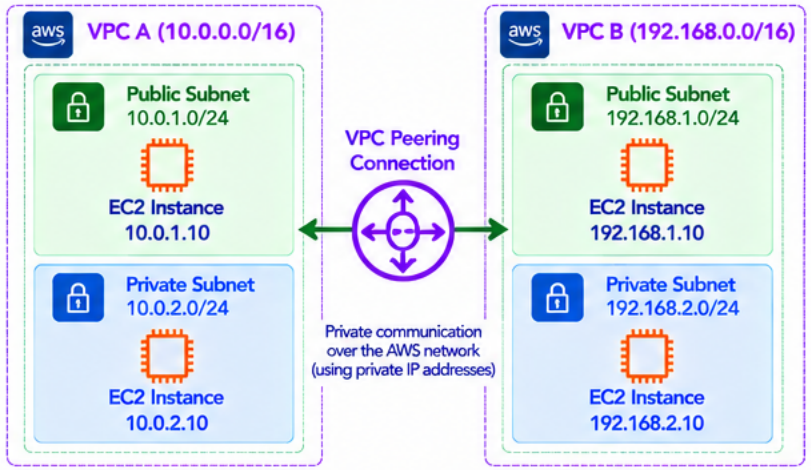
What is VPC Peering?

VPC Peering is a networking connection between two VPCs that enables them to route traffic to each other using private IP addresses over the AWS network. It does not use the internet, NAT, or VPN.

- Private and secure connection within AWS
- Low latency and high bandwidth
- Simple to set up
- Supports communication between resources in different VPCs
- Can be between VPCs in the same or different AWS accounts
- Transitive peering is not supported (e.g., A ↔ B and B ↔ C does not mean A ↔ C)

How VPC Peering Works (Same Region Example)



Route table in VPC A

Add route to 192.168.0.0/16 via VPC Peering

Route table in VPC B

Add route to 10.0.0.0/16 via VPC Peering

Key Points

- **Uses private IP addresses** — Traffic stays on the AWS network.
- **Non-transitive** — You must create direct peering between each pair of VPCs.
- **Same or different AWS accounts** — You can peer VPCs within the same account or across different accounts.
- **Same or different regions** — Supports intra-region and inter-region peering (inter-region uses the AWS global network).
- **Update route tables** — You must add routes in both VPC route tables.
- **No overlapping CIDR blocks** — VPCs with overlapping CIDR ranges cannot be peered.

Common Use Cases

- Connect development, staging and production environments
- Share databases or internal services across VPCs
- Connect VPCs in different AWS accounts (e.g., central network and application accounts)
- Enable communication between shared services (e.g., monitoring, logging)

Route Table Examples

VPC A Route Table (for 10.0.1.0/24)

Destination	Target
10.0.0.0/16	local
192.168.0.0/16	pcx-0abc123def456 (VPC Peering)
0.0.0.0/0	igw-xxxx (if internet access is needed)

VPC B Route Table (for 192.168.1.0/24)

Destination	Target
192.168.0.0/16	local
10.0.0.0/16	pcx-0abc123def456 (VPC Peering)
0.0.0.0/0	igw-xxxx (if internet access is needed)

Key Takeaways

- VPC Peering enables private connectivity between two VPCs.
- Update route tables on both sides.
- Ensure CIDR ranges do not overlap.
- Can be used across accounts and regions.
- Not transitive (use Transit Gateway for complex architectures).
- Keep security groups and network ACLs properly configured.

Limitations

- No transitive peering
- Cannot peer VPCs with overlapping CIDR blocks
- You must manage route tables and security groups
- Does not provide internet access (use IGW or NAT if needed)
- Not a replacement for AWS Transit Gateway for large-scale hub-and-spoke architectures

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AWS VPC Endpoints

Access AWS services privately from your VPC

VPC Endpoints allow your resources in a VPC to connect to supported AWS services using the AWS private network without going through the internet.

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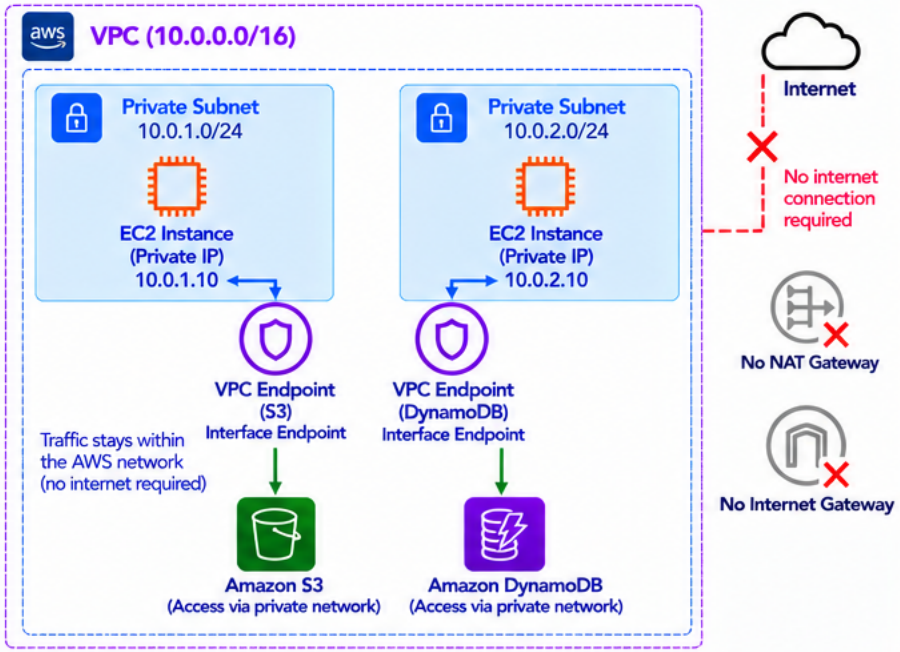
What is a VPC Endpoint?

A VPC Endpoint enables private connectivity between your VPC and supported AWS services. It allows your instances to access AWS services using private IP addresses without using the internet, NAT Gateway or VPN.

Key Benefits

- Access AWS services privately
- No need for internet, NAT or VPN
- Improves security (keeps traffic within AWS network)
- Reduces data transfer costs
- Supports fine-grained access using IAM and resource policies
- Easy to set up and manage

How VPC Endpoints Work (Example: Access S3 and DynamoDB)



Types of VPC Endpoints

Interface Endpoint

- Uses an elastic network interface (ENI) in your subnet
- Provides a private IP address
- Supports most AWS services (e.g., S3, DynamoDB, EC2, ECR, CloudWatch, etc.)
- Charged per hour and per GB of data

Gateway Endpoint

- Used for S3 and DynamoDB only
- Adds a route to your route table
- Uses a prefix list (no ENI)
- No hourly charge (free)
- Highly available and scales automatically

Common Use Cases

- Access Amazon S3 privately (e.g., backups, data lakes)
- Access DynamoDB privately
- Access AWS services from private subnets (without NAT or internet)
- Improve security and meet compliance requirements
- Access AWS services across multiple VPCs (using PrivateLink)

Route Table (Gateway Endpoint Example for S3)

Destination	Target	Purpose
10.0.0.0/16	local	VPC local traffic
pl-63a5400a	vpce-0abc123456789	Route to S3 via Gateway Endpoint
0.0.0.0/0	no route (or NAT/IGW for other traffic)	No internet required for S3 access

Note: For interface endpoints, no route table change is needed. DNS resolves the service to the private IP of the endpoint.

DNS Resolution (Interface Endpoint)

When you use the AWS service domain name (e.g., s3.amazonaws.com), DNS automatically resolves it to the private IP address of the VPC endpoint within your VPC.

Your application can use the same service endpoint name without any code changes.

Key Takeaways

- VPC Endpoints enable private access to AWS services.
- Choose Interface or Gateway endpoints based on the service.
- No internet, NAT or VPN is required.
- Improve security, reduce costs and simplify architecture.
- Use proper IAM and endpoint policies to control access.